The **Mobile App** Project Canvas (ver. 1.0)

Project Name
GREEN MILES

CONCEPT
Describe the idea of the app you want to create

Turns everyday journeys into measurable climate action. The user picks a transport mode (walk, cycle, motorbike, bus, petrol car), taps Start Trip, and GPS tracks the route live. Every trip is scored against a petrol-car baseline of 0.170 kg CO2/km, so the saving is a real number. Savings roll up into a weekly dashboard, a statistics screen, a leaderboard and a rewards catalogue.

OBJECTIVES & PURPOSES
What goal do you want to achieve with this project?
Remember: it must be measurable and shared with the team

Auto-compute CO2 saved on 100% of completed trips — zero manual data entry. Drive users to a personal weekly CO2 goal. Trips taken offline sync successfully once connectivity returns. Keep leaderboard data trustworthy: anti-cheat cancels implausible trips before they are saved. Ship an installable Android APK as the MVP.

VALUES
What creates a real value for your user and which ties him to your product?
Focus on max 3 values and focus your app around themes.
Delete what doesn't create value.

Visible impact - Abstract 'carbon' becomes a concrete number per trip, per day, per week.
Earned trustworthy numbers - Speed caps, stationary detection and server-side totals mean a rank is genuinely earned.
Compete without exposure - Friendly weekly / monthly / all-time competition, with a one-tap switch to appear as 'Private User'.

PERSONAS
Describe the archetype of the users who will use your app.

Maya — eco commuter
Already walks and cycles. Wants proof her habits matter.
Dan — student
Chases ranks, streaks and rewards more than climate guilt.
Priya — default driver
Drives from habit. Switches to the bus when shown the exact saving.
Sam — privacy-minded user
Will compete, but not under a full name. Uses the private toggle.
Catalogue admin (back office)
Curates rewards in Supabase, never opens the app

PROBLEMS TO SOLVE
Weaknesses of the current / similar solution. What is not working?
What are the critical points?

The carbon cost of daily travel is invisible. There is no feedback loop between a choice and its impact. Existing trackers are fitness-first (steps, calories) and never compare one mode against another. On stock Android, Doze silences GPS minutes after the screen turns off — trips come back as a start point and an end point. Lose signal, lose the trip. Most apps have no offline queue. Self-reported 'green' apps are trivial to fake, so their leaderboards are meaningless. Carbon calculators are one-off web forms; nothing accumulates or persists. Users must choose between competing publicly and staying anonymous.

COMPONENTS
What are the main modules that make up the app?
Put all the features in a well defined component.

Modules: Onboarding & Auth, Trip Tracking, Dashboard, Statistics, Ranking, Marketplace, Inbox, Profile & Preferences
Core services: LocalNotifier, NetworkWatcher, PendingTripStore, BatteryUnrestrict, BrandTheme (design system & strings).
Backend (Supabase): profiles / trips / rewards / inbox / user_preferences, row-level security on every table, signup and post-trip triggers, public avatars bucket.

FEATURES
What the app should do? What services are provided?
List all the features your users expect to find
Collect all ideas and then select those that allow you to better reach the goal.

Auth & onboarding
• Three-screen intro carousel; email + password sign-up with a strength checklist (8+ chars, upper, lower, digit, symbol).
• Email verification, OTP password reset, change email, change password, account deletion.
Trip tracking — the core loop
• Mode picker: Walking, Cycling, Motorbike, Bus, Petrol Car, each with its own emission factor.
• Live map with drawn route, distance, duration, smoothed live speed and running CO2 saved.
• Anti-cheat: per-mode speed caps (walk 12, bike 20, bus 80, motorbike 100, car 120 km/h), 15 s grace window, up to 3 warnings, then cancellation. Stationary trips warned at 45 s, cancelled at 1 min.
• GPS noise handling: accuracy gates, exponential-moving-average speed smoothing, implausible-sample rejection.
• Persistent foreground notification while a trip runs; prompt to disable battery optimisation.
• Offline queue — trips recorded without connectivity are stored locally and pushed on reconnect.
Impact, competition & retention
• Dashboard: weekly CO2 dial scaled to the user's goal, plus a 7-day bar chart of daily savings.
• Statistics: weekly total, average per day, active days, best day, transport-mode breakdown, trend.
• Leaderboard with Weekly / Monthly / All-Time windows and a podium for the top three.
• Rewards marketplace.
• Inbox for system messages: trip recorded, trip cancelled, welcome.
• Profile: avatar upload, lifetime totals (distance, trips, CO2), full trip history.
• Preferences: weekly CO2 goal, public/private leaderboard visibility, password change.

MILESTONES
Define the roadmap of the project.
Start from the MVP and then scale.

1 - Foundation
Flutter scaffold, MVVM + Provider, brand theme, Supabase schema + RLS, seed data.
2 - MVP
Auth, mode picker, live GPS tracking, CO2 calculation, trip persistence, dashboard.
3 - Engagement
Statistics, leaderboard windows, inbox, profile & trip history, preferences.
4 - Trust & reliability
Anti-cheat caps and stationary detection, offline queue, foreground notification.
5 - Release
Rewards catalogue, release keystore, real app ID, store listing, privacy policy.
6 - Scale (post-MVP)
Reward redemption, team challenges, iOS build, regional factors, automated tests.

STAKEHOLDERS
Who need to be involved in the project? What is their role and who makes the decisions?
In Enterprise Projects the stakeholders can be many, beware!

• Project owner / lead developer — sole committer; owns scope and the final call.
• End users — commuters, students, casual switchers.
• Supabase — Database provider (auth, Postgres, storage).
• OpenStreetMap — map tiles provider.
• Reward partners / sponsors — supply the catalogue.
• Google Play — for background location.
• Supervisor.

RISKS
Every project has risks, knowing which at the beginning allows to mitigating them. What could go wrong and not help achieve the goal?

• OEM battery killers stop GPS mid-trip.
• Users deny 'Allow all the time' location — the core loop cannot run.
• Play Store rejection: background location needs justification.
• Anti-cheat false positives (tunnels, GPS jumps, e-bikes) cancel honest trips.
• Emission factors are hard-coded and region-agnostic.
• .env secrets must never reach the repo.
• Single-vendor dependency on the Supabase free tier.
• Leaderboard shows names by default: privacy / GDPR.

DELIVERABLES
What will the team have to produce?
In addition to the app you could have a UI prototype, a middleware, a launch plan, documentation ...

• Flutter Android app (source + signed release APK).
• Supabase package: schema.sql, seed.sql, delete_seed.sql.
• Design system: BrandTheme, launcher/adaptive icons, logo.
• UI prototype / screen flows for the screens.
• README setup guide (Supabase, .env, build commands).
• Privacy policy and data-deletion statement.
• Test plan: anti-cheat, offline sync, permission-denied paths.
• Launch plan and store listing assets.

APP NAME
:)

Green Miles

CONTEXT OF USAGE & COVERAGE
Where will the abb be used?
With full WiFi/4G signal or offline mode?

Outdoors and in motion,Needs background GPS; offline trips queue and sync later.

TECHNOLOGY
Native, HTML, Hybrid, ...

Native-compiled Flutter / Dart, MVVM with Provider. Supabase. flutter_map + OSM, fl_chart, local notifications.

PLATFORM, OS, ...
Which device and operating system?

Android phones, compile/target SDK 31+, Java 17.

ORIENTATION
Portrait, Landscape or both

Portrait — every screen is designed portrait-first. No orientation lock is set in code, so landscape is untested.

RELEASE
Public Marketplace or Enterprise App Store?

Public marketplace (Google Play).